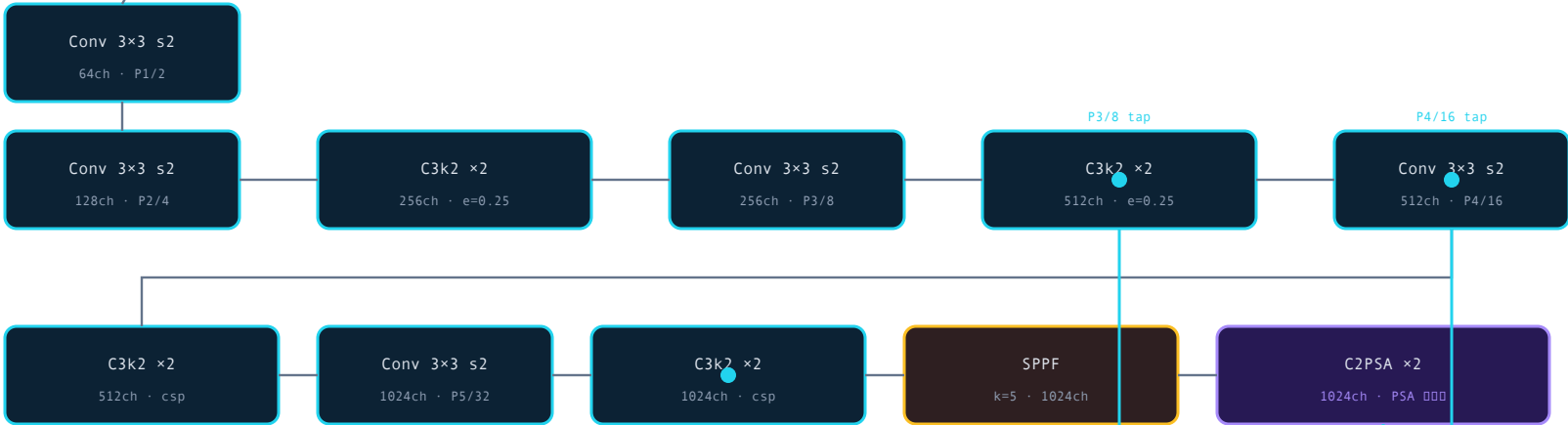


INPUT

RGB Image
640 × 640 × 3

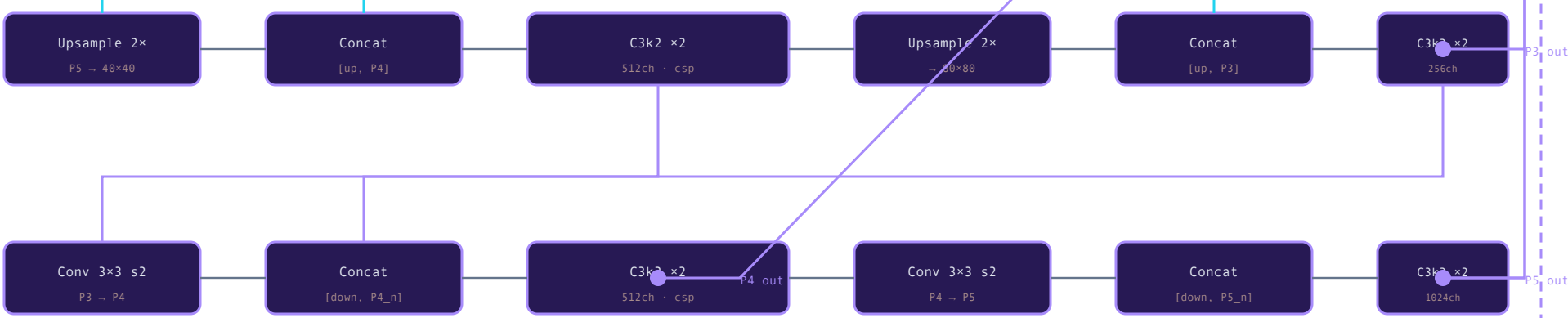
BACKBONE · C3k2 + C2PSA + SPPF

P1/2 → P5/32



NECK · PANet (FPN top-down + PAN bottom-up)

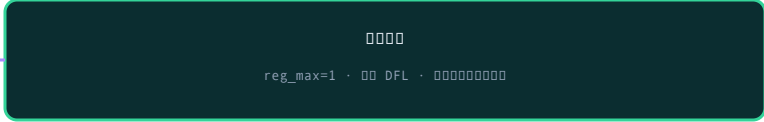
00000000



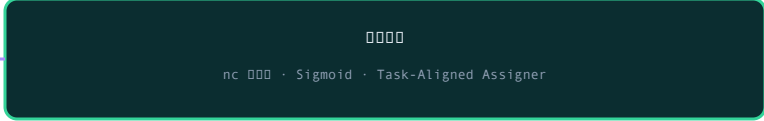
HEAD · Segment26 (DFL-free · 00)

DFL-free · e2e default

BOX REGRESSION (DFL-free)



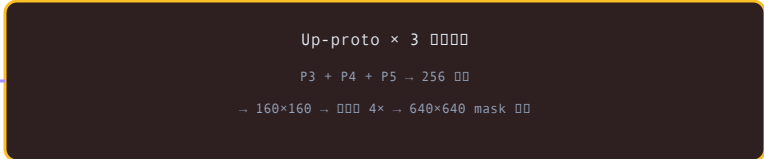
CLASSIFICATION (BCE)



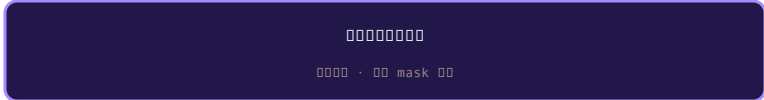
MASK COEFFICIENTS (32 0)



PROTO MODULE (000 · 000)



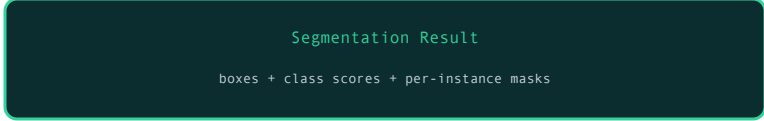
SEMANTIC SEG AUX LOSS



DUAL DETECTION HEAD



OUTPUT



YOLO26s-seg · SPEC SHEET

00	Instance Segmentation · 0000	0 0000	DFL-free box regression
00	640 × 640 × 3 (RGB)	0 0000	NMS-free end-to-end
Backbone	Conv + 4× C3k2 + SPPF + C2PSA	0 0000	MuSGD 000
Neck	PANet (FPN top-down + PAN bottom-up)	0 0000	STAL 000000
Head	Segment26 · DFL-free · 0000 (e2e + NMS)	0 0000	Up-proto 000
00 / 000	309 0 · 11.5M params (s) · 37.4 GFLOPs	0 0000	Sem seg aux loss
COCO mAP (seg)	box +2.5 AP · mask +3.7 AP (vs YOLO11)	0 00	arXiv:2606.03748 (2026)
0000	T4 TensorRT 2.5ms · CPU ONNX 87ms	0 00	github.com/ultralytics/ultralytics

TRAINING RECIPE

- MuSGD Optimizer
- SGD + Muon 00 · LLM 0000000 CV
- Progressive Loss
- 00000000 1-to-many 0 → 00 1-to-1 0
- STAL (Small-Target Aware Label)
- 000000000000000 · 000/000000
- Semantic Seg Aux Loss
- 0000000000 · +2.5 box / +3.7 mask AP (vs YOLO11)